

POLS 641

Introductory Political Analysis

Class 4 and Probably 5: Measurement

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Warm-Up

You want to know what impact the online vs in-person classes has on student performance here at NIU.

How could make the causal claim that the move online decreases student performance?

What about an observational study?

What would this look like using core model notation?

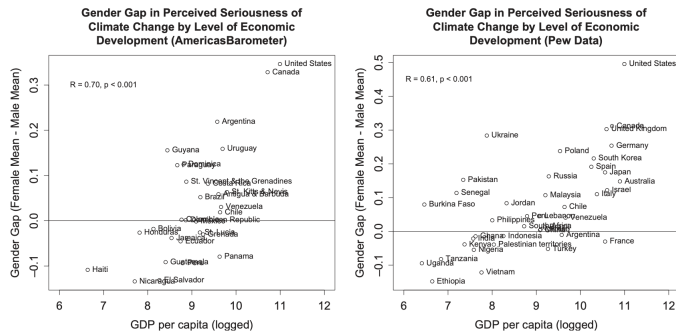
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Today's(and next week's) Agenda

- RiA: Facing Change
- Housekeeping
- The Statistical Inference Process
- Review Probability
- Random Variables
- Measurement (next week)
 - Validity
 - Reliability
- Describing Variables
- The Statistical Inference Process Revisited

An Example

FIGURE 1. The Gender Gap in Climate Concern by Country and Level of Economic Development, 2015–2017



Note: Data are from the AmericasBarometer (*left panel*) and Pew (*right panel*), with climate concern standardized so that the mean value is 0 and the standard deviation is 1. The values are highly correlated for both surveys. AmericasBarometer: $\rho = 0.70, p < 0.001$. Pew: $\rho = 0.61, p < 0.001$.

An Example: Quantitative Methods in Action

- What is the puzzle?
- What methods are used to address their research question?
- What are they explaining? How do they measure it? (DV)
- What is doing the explaining? How do they measure it? (IV)

An Example

Regression table

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(AmericasBarometer)			(Pew)		
(Intercept)	0.996* (0.463)	0.641 (0.395)	0.114 (0.347)	0.858* (0.352)	3.170* (1.325)	3.244* (1.367)
Women	-0.880*** (0.095)	-0.999*** (0.102)	-0.522*** (0.141)	-0.432*** (0.061)	-0.612* (0.279)	-0.632* (0.282)
GDPpc	-0.108* (0.052)	-0.045 (0.043)	-0.031 (0.037)	-0.098* (0.038)	-0.298* (0.132)	-0.311* (0.136)
Women $\times$ GDPpc	0.101*** (0.011)	0.114*** (0.011)	0.060*** (0.016)	0.057*** (0.007)	0.080** (0.028)	0.075** (0.028)
Ideology		-0.028*** (0.002)	-0.012*** (0.002)		-0.048*** (0.006)	-0.048*** (0.006)
Education			0.026*** (0.002)			0.011** (0.002)
Household income			0.006*** (0.001)			0.011** (0.001)
Log Likelihood	-53,871	-43,943	-32,687	-59,789	-11,595	-11,292
N obs.	38,780	31,628	24,619	44,099	9,202	8,982
N countries	29	22	20	40	12	12

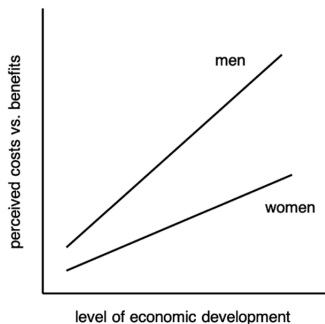
An Example: Quantitative Methods in Action

- How do the authors demonstrate the robustness of their findings?
- What is the benefit of using focus groups?
- What are the key methodological terms you don't understand?
- Are you convinced?

An Example

Visualizing the theory

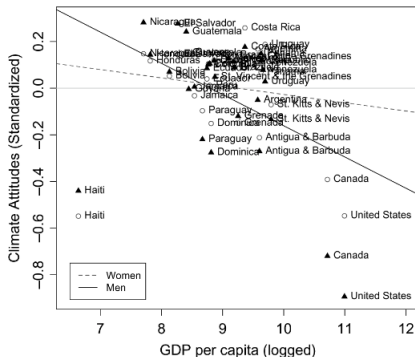
FIGURE 3. The Perceived Costs versus Benefits of Climate Change Mitigation Depend on a Country's Level of Economic Development and an Individual's Gender



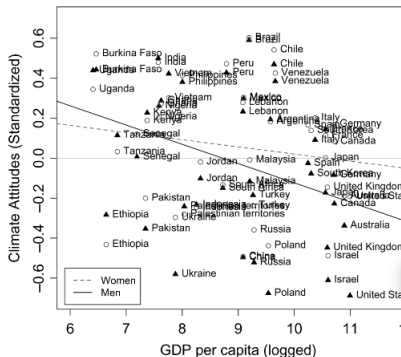
Note: As the costs become larger relative to the benefits, individual concern about climate change will decrease.

An Example

Average Perceived Seriousness of Climate Change by Level of Economic Development (AmericasBarometer)



Average Perceived Seriousness of Climate Change by Level of Economic Development (Pew)

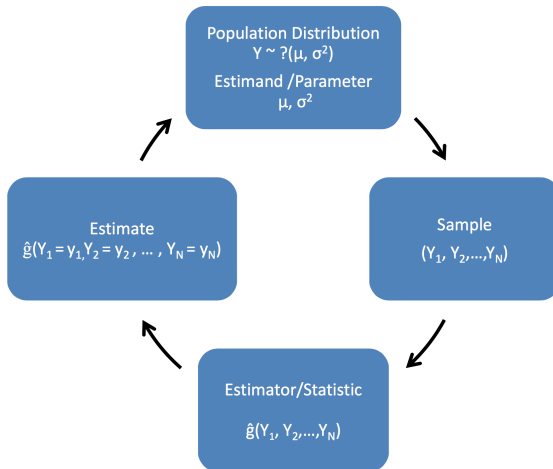


Housekeeping

- ① Problem Set #1 - Due tomorrow
- ② Problem Set #2 - Available Friday, Due in two weeks
- ③ Your Final Project - Research Questions due October 2nd
- ④ Readings for Next Week:
 - Adcock and Collier
 - Read 1, Skim 1: Pollock or Imai

Key Idea

The statistical inference process.



Some Definitions

- Estimator: Concept or relationship in the population which is to be estimated in a statistical analysis
- Estimator: A rule for calculating an estimate of a given quantity based on observed data Function of the observations, i.e., how observations are put together
- Estimation: The process of finding an estimate, or approximation, which is a value that is usable for some purpose even if input data may be incomplete, uncertain, or unstable (value derived from the best information available)
- Estimate: The actual derived value of the concept or relationship

This language is confusing, but the underlying idea is very important

Examples

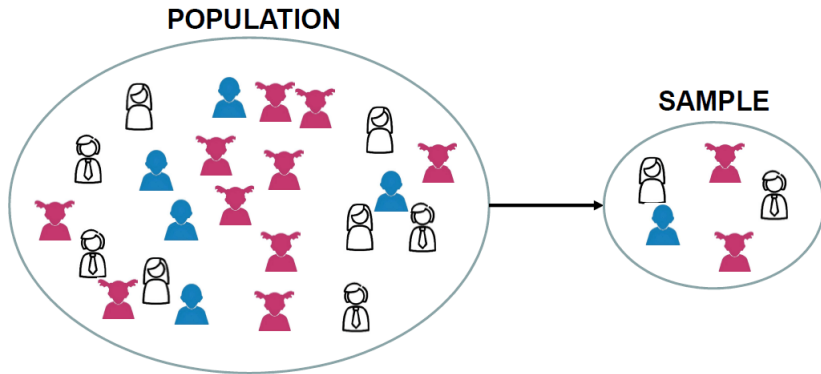
- Example 1: To measure American's support for climate change, our estimand, we take a random sample of the US population and find the mean of a specific question, the estimator, to give us an estimate.
- Example 2: We rely on the observations in our sample and use a linear (regression) function (the estimator) to estimate the causal effect of education on income in our sample which is our estimate for the population-level causal effect (the estimand).

We can use these terms for both a single variable or the relationship between multiple variables (e.g. OLS regression as the estimator of the relationship between X and Y)

Population vs Sample

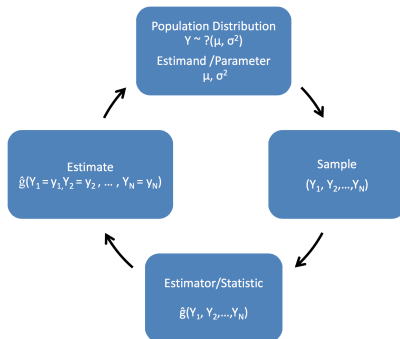
Population	The overall collection of individuals, beyond just the sample
Sample	The collection of individuals on which statistical analyses are performed, and from which general trends for the <i>population</i> are inferred
Individual	Also called an 'object' or 'unit,' a single data point contributing to the sample

Population and Sample



Key Idea

The statistical inference process.



How my brain thinks about it: Helps us think through what is "true," but unobservable, and how that leads to a probabilistic understanding of our best guess

Review: Probability

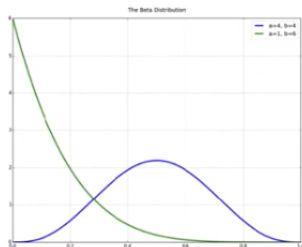
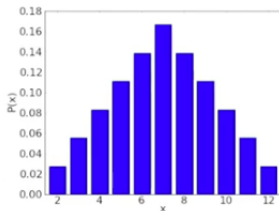
- Necessary when analyzing situations characterized by "chance" or "randomness" i.e. data-generating process that can realize any number of different values.
- As social scientist, we tend see the world in probabilistic rather than deterministic ways
- ie our outcomes of interest are probabilistic

Big Ideas

- an understanding of probability theory helps us understand how what we observe relates to the real world.
- statistics has to do with understanding properties of the distribution of potential outcomes, not just the data you have
- idea that you can observe different outcomes with different probabilities, even when you only observe one outcome

Probability Density Function

- The outcomes listed must be disjoint.
- Each probability must be between 0 and 1.
- The probabilities must total 1.



Random Variable

A random variable X is a (real-valued) function that maps the sample space to the real numbers (i.e. the function assigns a numerical value to each possible outcome of the experiment)

a way to think about outcomes of random processes to numbers

Quick Clarification(again): Randomness vs. Randomization

- The textbook uses both terms. It's a bit confusing.
- Important: These two terms do not mean the same thing.
 - Randomization (aka random assignment): using a coin toss to create treatment and control groups; creates exogeneity.
 - Randomness: noise in the data (i.e. effects from sampling on a given day) - could go away with larger sample sizes. We will address some of these concerns with t-tests (p-values) and confidence intervals. Can differentiate between sampling randomness and modeled randomness.
 - **Random Variable** A variable that takes on values in a range and with the probabilities defined by the distribution

Goal: Understanding the ‘Distribution of the mean’

Two related goals:

- Goal 1: Understand what is meant by the “*distribution of the mean.*”
- Goal 2: Understand how to figure out the distribution of a mean in theory, given you only have a *sample of individuals* and can’t actually see the *distribution of the mean.*

The first part we can see conceptually from simulations and graphs.

The second part will require some statistical theory.
We will revisit this again in our R workshop this week

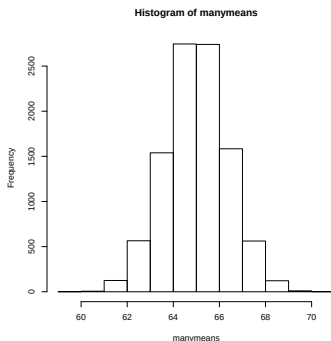
Part 1: Distribution of the mean, conceptually

Here we convey the concept of a distribution of the mean

Suppose you can draw (sample) some quantity as many times as you want:

- Example: You gather data on the height of people coming down the hall, pretending you never run out of people. All the people are from the general NIU population.
- Take the mean (average) of, say, 20 peoples' height
- Do that over and over again.

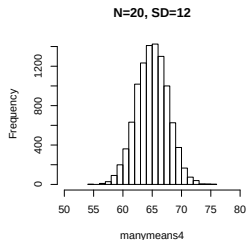
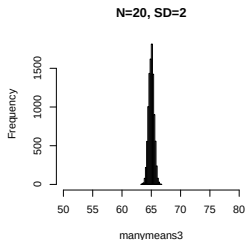
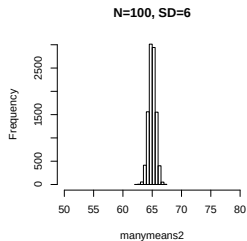
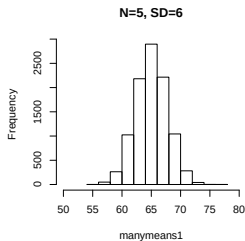
Part 1: Distribution of the mean, conceptually



How does this change if we change the number of people we are averaging together ($N = 5, 20$ or 100)?

How do you think this changes if the natural variation of height (sd) in the population was much larger or much smaller ($sd = 2, 6$, or 12)?

Part 1: Distribution of the mean, conceptually



What to note about distribution of the mean

Remember, the mean you get from a given sample is one number, but we're talking about the distribution governing what it *could have* looked like.

When working with *actual data*, you only see the mean once, and you can't do this simulation exercise directly. (You can't collect the data over and over again – you only have one sample in the data you have).

But, it turns out that the *only* thing the *distribution of the mean* depends on is N and the mean and variance (SD^2) of the individuals

- this will allow us to construct null distributions involving the mean very easily without simulation
- it will also give us “confidence intervals” telling us how different the mean could have been if we had a slightly different sample (because of noise).
- Two quantities: based only on N and estimated mean and variance of the underlying population.

Why Random Variables (RVs)

We get a new piece of information (e.g. flip a coin, roll a dice, or measure someone's height). This is called:

- a random draw, or
- a random experiment

For any random experiment, the thing we are measuring is called a “**random variable**”.

Key properties of random variables:

- It has “**a set**” of possible values it can take (e.g. $\{heads, tails\}$, $\{1, \dots, 6\}$, or a positive number).
- Those outcomes have different probabilities of occurring. These are characterized by a distribution.
- You don't know which of those it will take until you do the random experiment/draw/observing.

Expectation and Variance

Again, random variables are measurements drawn from a distribution, whether we know that distribution or not.

Two important features of any distributions are **expectation** and **variance**.

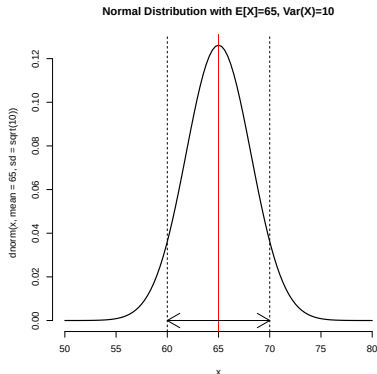
- Expectation: the best guess about what number will be drawn from the distribution
- Variance: how far the numbers you draw tend to be from that best guess.

We will analyze the expectation and variance for a number of distributions, starting with normal.

A Normally Distributed Variable

Say we are interested in the height of the next person to walk through the door.

We will assume this random variable is distributed normally with expectation 65 inches and variance of 10 inches:



A bit more on expectation and variance The expectation, or $E[X]$ for a random variable,

- is a different concept from the average or mean, because it is something that exists for any random variable, not something you get from the data.
- but, you can think of it loosely as the “average” of what you could get from the distribution
- if you could draw repeatedly from a distribution and take an average, it would get closer and closer to the expectation

Final words on normal distributions

In two weeks we'll see why the normal distribution appears so often.

For now, you should know that a normal distribution is completely described by *only* its expectation and variance.

Shift to thinking in terms of mean **AND** variation

Samples as Collections of Random Variables

Normally we have a *sample*: multiple RVs all with the same distribution. E.g.

- person 1 has $height_1$, a RV drawn from $N(\mu, \sigma^2)$
- person 2 has $height_2$, a separate RV drawn from $N(\mu, \sigma^2)$
- ...
- person N has $height_N$, yet another RV drawn from $N(\mu, \sigma^2)$

Even though each observation is a different RV, since they all come from the same distribution, we leverage the sample to learn something about that distribution.

The Sample Mean

The sample mean of $X_1, X_2, \dots, X_N$ is:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i = \frac{1}{N} [X_1 + X_2 + \dots + X_N]$$

Important notes:

- This is an example of an *estimator*: a function you can compute with data. You can use these estimators in R, for example `mean(object)`.
- **The mean is itself a random variable!** It takes on some particular value in your experiment, but it could have taken different values
- We are trying to understand what the *distribution of the mean* must look like, despite seeing the mean only once.

Why have we done all this?!

Remember, the big idea was that even when we only observe a mean once we can say a lot about how the mean would be *distributed* (as if we could observe it over and over again).

huge part of the statistical machinery depends on this finding. In two we'll see how this allows us to:

- Construct “confidence intervals” around your estimated mean to characterize our uncertainty
- hypothesis tests, get p-values, etc. for one-sample and two-samples tests involving means and eventually stars
- Other hypothesis tests for categorical outcomes, binary outcomes, counts, etc. will follow similar patterns.

Why have we done all this?!

- Takes on values in a range and with the probabilities defined by the distribution
- A random variable X is a (real-valued) function that maps the sample space to the real numbers (i.e. the function assigns a numerical value to each possible outcome of the experiment)
 - We define a random variable by its **Expectation** and **Variation**
 - $E[X]$ is almost always measured as the mean Variance is almost always measured as the standard deviation(more on this later)

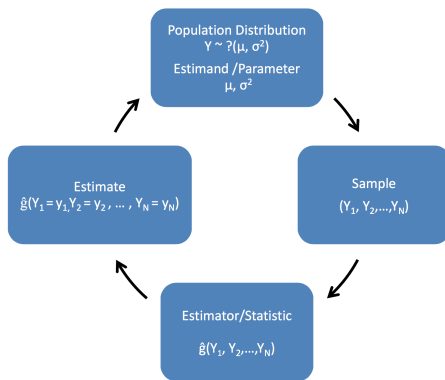
On a final note, this is tough stuff but paves the way for much of what we will do, so go over these slides.

Questions?

New Key Terms(So Far)

- 1 Population
- 2 Sample
- 3 Random Variable
- 4 Distribution(probability distribution)
- 5 Probability Density Function
- 6 Normal Distribution
- 7 Estimand
- 8 Estimator
- 9 Estimate
- 10 Expectation

Statistical Inference Process



How do we design research to use the best estimator?

Next time we will shift our focus from the math to thinking about concepts. ie estimators



estimand

Ingredients

150g unsalted butter, plus extra for greasing

150g plain chocolate, broken into pieces

150g plain flour

1/2 tsp baking powder

1/2 tsp bicarbonate of soda

200g light muscovado sugar

2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.

estimator



estimate

The things you find on the internet

Part 2: Operationalization and Describing Single Variables

Agenda

- Concept to Measurement
- Unit of Analysis
- Reliability and validity
- Descriptive Statistics
- Measures of Central Tendency
- Variation

Concepts

- Our description (and inference) is based on accurately measuring the concept
- Concept -> Operationalization -> Measure
- What are some examples of key (contested?) concepts in political science?

Key Elements of Concepts

- must be concrete
- must vary
- what are some “concepts” you are interested in?
- identifying new concepts and describing their variation is a great way to start a research project!

Key Elements of Concepts

- A conceptual definition clearly describes the concepts measurable properties and specifies the units of analysis to which the concept applies
 - specifies units of analysis
 - suggests the potential for variation along a constitutive set of characteristics
 - provides a theoretical way of determining the degree to which units exhibit those characteristics (i.e. the concept must be measurable in theory and hopefully in practice)
- The challenge for us is being able to quantify complex concepts